Modeling Cell-Cell Interactions in T-Cell Dependent Cytotoxicity (TDCC) Based Cancer Therapeutics

Project Takeda Oncology

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1. Executive Summary

In this project, we developed a multi-layered modeling framework to better understand T-cell dependent cytotoxicity (TDCC), with a focus on CAR T-cell-based therapies. By combining Agent-Based Modeling (ABM), systems of Ordinary Differential Equations (ODEs), and Master Equation (ME) approaches, we aimed to link in vitro assay results to mechanistic insights and improve predictive modeling of immune-tumor dynamics.

A central finding from our simulations is that the **effector-to-target (E:T) ratio** alone does not fully explain treatment outcomes. While widely used in preclinical design, our models show that **absolute cell counts** also play a critical role. At fixed E:T ratios, wells with higher total cell numbers consistently exhibited stronger cytotoxic responses, both in ABM and ODE simulations. This suggests that spatial density and interaction likelihoods – especially in low-count regimes – can shape therapeutic effectiveness in ways not captured by ratios alone.

We validated our models using both synthetic and experimental data. The ABM and extended ODE model produced consistent trends when fit to simulated trajectories. However, when applied to sparse endpoint-only lab data (18 samples), both models struggled to match outcomes. To complement these limitations, we turned to a Hill-type empirical model, which despite its simplicity, achieved strong fits and highlighted saturation effects and steep response dynamics.

Our exploration of the Master Equation offered a probabilistic lens on system behavior, though its utility in this specific setting was limited by smooth distributions and lack of rare-event structure. Nevertheless, it remains promising for future studies involving early stochastic behavior or treatment escape.

This work provides Takeda with a flexible, interpretable toolkit for modeling TDCC systems, and lays the foundation for future extensions incorporating effector exhaustion, time-series data, and refined spatial mechanisms.

2. Introduction

2.1 Biological Motivation

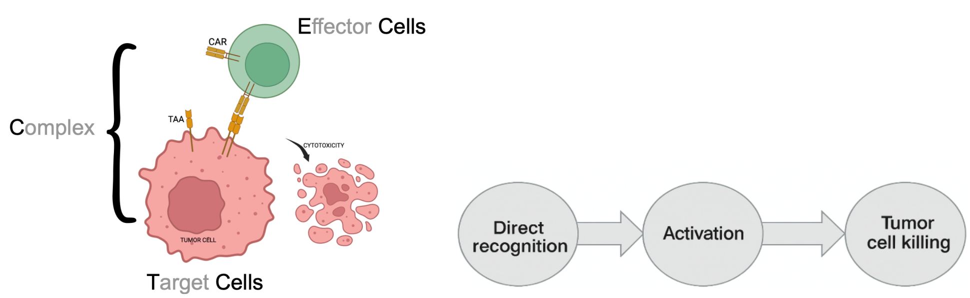
Immunotherapies that harness the immune system to selectively eliminate cancer cells have transformed cancer treatment. Among the most potent approaches are T cell-dependent cytotoxicity (TDCC) therapies, such as Chimeric Antigen Receptor (CAR) T cells. These therapies rely on engineered immune cells to recognize and destroy tumor cells that express tumor-associated antigens (TAAs) on their surface.

To better understand the mechanisms behind these therapies and improve their predictive power, we develop quantitative models that capture how effector and tumor cells interact over time. Our goal is to move beyond qualitative descriptions and build interpretable tools that link experimental data to underlying biological processes.

2.2 Mechanism of CAR T Cells and Problem Introduction

At a mechanistic level, CAR T cells are designed to recognize TAAs displayed on the surface of cancer cells. This recognition is mediated by the CAR molecule - a synthetic receptor that binds a specific antigen. Once bound, the CAR T cell becomes activated and releases cytotoxic agents that kill the target cell. This recognition → activation → killing sequence forms the biological foundation of TDCC therapies.

As shown below, this interaction typically involves the formation of a transient complex between an effector cell and a tumor cell, culminating in target cell death. While this mechanism is well established, optimizing therapeutic design and predicting efficacy requires moving beyond qualitative descriptions to quantitative models that capture these dynamics under varying conditions.



One of the most commonly reported variables in cytotoxicity assays is the effector-to-target (E:T) ratio – the number of effector cells relative to tumor cells. In vitro assays are often conducted at high E:T ratios (e.g., 5:1, 10:1, 20:1) to ensure robust cytotoxicity signals. However, these conditions may not reflect those found in vivo, where E:T ratios can drop below 1:10 or even 1:100, especially in solid tumors.

This raises several important modeling questions:

- Does the E:T ratio significantly affect therapeutic outcomes?
- Are high-ratio assay results reliable predictors of clinical performance?
- What biological factors - such as spatial organization, timing, or stochasticity - drive variability in cytotoxic effectiveness?

In the following section, we introduce the models we developed to answer these questions.

3. Modeling Framework

We use two modeling approaches to study TDCC dynamics: a stochastic agent-based model (ABM) that simulates individual cell interactions in space, and a deterministic ordinary differential equation (ODE) model that captures average population behavior. Before introducing these models, we first define the F-metric, a simple measure of treatment effectiveness based on tumor cell reduction.

3.1 F-Metric

To quantify treatment effectiveness across different scenarios, we define a simple outcome measure called the **F-metric**. For a given initial condition, we simulate two versions of the system: one with effector cells (treatment) and one without (control). The F-metric is then defined as:

$$F(t) = \frac{T(t, E_0)}{T(t, 0)} = \frac{\text{Target Left with Treatment}}{\text{Target Left without Treatment}}.$$

An F value close to 0 indicates strong cytotoxic effect, while a value near 1 implies little to no impact. This relative measure allows us to compare treatment efficacy across different initial conditions, E:T ratios, and modeling frameworks in a consistent and interpretable way.

3.2 Agent-Based Model (ABM)

The agent-based model (ABM) simulates individual effector (E) and target (T) cells in space, capturing stochastic behaviors and local interactions that are difficult to represent in population-level models.

In our setup, a specified number of E and T cells, each with radius r, are placed uniformly at random in a square well of side length L. At each time step, E cells move via a Gaussian random walk with standard deviation m. When an E cell comes into contact with a T cell, it has a probability k_{on} (binding rate) of binding and forming a complex (C cell).

Each C cell evolves independently:

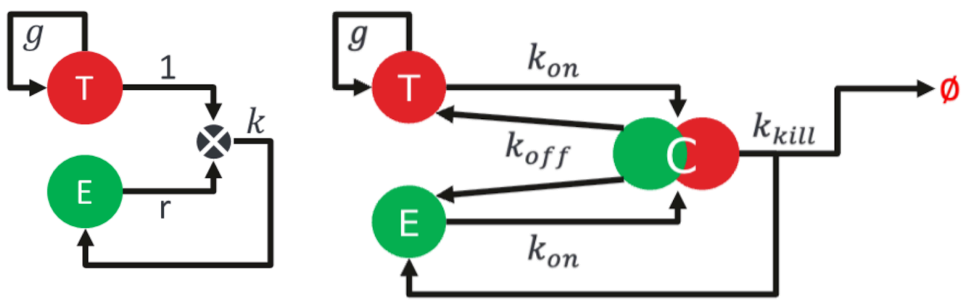
- with probability k_{off} (unbinding rate), it unbinds back into an E and a T cell;
- with probability k_{kill}(killing rate), it kills the T cell and returns the E cell;
- with remaining probability $(1 - k_{\text{off}} - k_{\text{kill}})$, the complex persists.

In addition to these interactions, E and T cells can undergo natural proliferation at rates g_E and g_T, respectively. To remain consistent with the ODE model introduced below, we typically set $g_E = 0$ and refer to g_T simply as g (growth rate).

3.3 Ordinary Differential Equation (ODE) Model

Dynamical systems modeling is a well-established approach in mathematical oncology. In particular, ordinary differential equations (ODEs) offer a compact and scalable framework for modeling the average behavior of large cell populations over time. In our context, ODEs describe the time evolution of effector cells E(t), target cells T(t), and their complexes C(t).

We consider two formulations: a simplified model based on population-level mass-action kinetics, and an extended system inspired by enzyme-substrate interactions, where E cells act as “enzymes” and T cells as “substrates.” These models are summarized schematically in Figure 1.



(a) (b)

Figure 1: (a) The life cycle diagram for the simplified model and (b) The life cycle diagram for the extended system

The first model resembles a predator-prey system:

$$\frac{dT}{dt} = gT - kE^rT \tag{1}$$

Here, g is the target cell growth rate, k is an effective killing rate, and r is a stoichiometric constant, often set to 1. This one-dimensional model is illustrated by the life cycle diagram in Figure 1 (a).

The second, more mechanistic model tracks effector-target binding and unbinding explicitly, as well as complex formation and resolution:

$$\begin{cases} \frac{dT}{dt} = gT - k_{\text{on}}E \cdot T + k_{\text{off}}C, \\ \frac{dE}{dT} = -k_{\text{on}}E \cdot T + k_{\text{off}}C + k_{\text{kill}}C, \\ \frac{dC}{dt} = k_{\text{on}}E \cdot T - k_{\text{off}}C - k_{\text{kill}}C, \end{cases} \tag{2}$$

where g is the intrinsic growth rate of target cells, k_{on} and k_{off} are the binding and unbinding rates, and k_{kill} is the rate at which bound complexes result in target cell death. This model is summarized in the life cycle diagram shown in Figure 1 (b).

Together, these ODE models capture the core dynamics of immune-mediated killing at the population level. The extended system can be further expanded to include additional biological mechanisms, such as cytokine signaling, drug response, or resistance pathways, making it a flexible tool for both theoretical analysis and data fitting. We refer to the extended three-equation system as "**the ODE model**" in the following discussion unless otherwise specified.

3.4 Pseudocode

This section formalizes the core modeling components of our framework. We describe, in pseudocode format:

- 3.4.1: An agent-based simulation that captures spatial and stochastic cell behaviors.
- 3.4.2: A corresponding ODE model for population-level dynamics.
- 3.4.3: A parameter fitting routine that enables data calibration of either model.

3.4.1 Agent-Based Modeling Simulation

```
class ABMSimulation:
    init(initial_conditions, parameters):
        E0 = number of initial E cells
        T0 = number of initial T cells
        initialize positions for E cells and T cells
        store all model parameters k_on, k_off, k_kill, g_E, g_T
    step():
        1. move all E cells via Brownian motion
        2. Binding: E + T → C
            for each E-T pair within binding distance:
                with probability k_on:
                    bind into a C complex
                    remove E and T from populations
                    add to C population
        3. C cell dynamics: unbinding, killing, or persists
            for each C cell:
                with probability k_off, unbind into E + T
                with probability k_kill, kill T, return E
                otherwise do nothing, C persists
        4. Cell natural growth
            with probability g_T, duplicate each T cell
            with probability g_E, duplicate each E cell
        5. Record
            record total counts, changes, and positions of E, T, C
```

```
    run(n_steps):
        for t in 0 to n_steps:
            call step(), break if stopping criteria met
        return records
```

3.4.2 ODE Model

```
class ODESimulation:
    init(initial_conditions, parameters):
        E0, T0, k_on, k_off, k_kill as in ABM simulation
        g = g_T in ABM simulation, g_E is assumed to be 0
    derivatives(t):
        dE = -k_on * E * T + k_off * C + k_kill * C
        dT = -k_on * E * T + k_off * C + g_T * T
        dC = +k_on * E * T - k_off * C - k_kill * C
        return [dE, dT, dC]
    run(time_span):
        integrate system over time span given initial E0 and T0
        return trajectories E(t), T(t), C(t)
```

3.4.3 Fit Data Using ABM or ODE

```
loss_function(parameters):
    run either ABMSimulation or ODESimulation with given parameters
    loss = difference between observed data and simulated data
    return loss
fit_model(data, param_bounds):
    data can be full time series or endpoints values only
    best_params = parameters within param_bounds with smallest loss
    return best_params
```

4. From Model to Data: Simulation, Fitting, and Interpretation

To evaluate how well different modeling approaches capture CAR T cell dynamics, we begin by using the ABM to generate synthetic data under biologically realistic assumptions. This allows us to test whether simpler population-level ODE models can reproduce the behavior observed in ABM simulations. We then use both ABM and ODE models to construct response surfaces and curves, enabling a quantitative comparison of how initial conditions and E:T ratios shape treatment effectiveness.

In the second part of this section, we turn to experimental data. Because the lab data consists of only endpoint measurements from 18 in vitro trials, we adapt our approach to fit final tumor cell counts instead of full trajectories. In addition to fitting ABM and ODE to this endpoint data, we also introduce an empirical Hill-type function. Together, these analyses help clarify what can and cannot be captured by mechanistic models when working with limited experimental input.

4.1 Fitting to Simulated Data

We begin by using the agent-based model (ABM) to simulate cytotoxicity under a biologically realistic setting. Each simulation runs for 96 steps, representing a 24-hour period with 15-minute intervals. The tumor (T) cell growth rate is set to 0.007, corresponding to a doubling time of 24 hours. Effector (E) cells move by a Gaussian random walk with a standard deviation of 0.02 in each time step, and the simulation takes place in a normalized 1×1 square well. The binding, unbinding, and killing probabilities are set to 0.4, 0.3, and 0.007, respectively.

Each simulation produces time series for the E, T, and C cell populations. To reduce the impact of stochastic fluctuations, we run each scenario 1000 times and take the median trajectory across runs. These smoothed trajectories serve as synthetic ground truth for evaluating how well simpler models can replicate ABM behavior. To explore generalizability, we simulate a grid of initial conditions where initial E and T cell counts each range from 100 to 10,000 in coarse steps (e.g., 100, 200, 300, …, 10,000), generating one synthetic trajectory per E-T pair. This grid allows us to test model performance across a wide range of cell densities and effector-to-target ratios.

We then fit both the simplified ODE model and the extended ODE model to these ABM-generated trajectories. As expected, the simplified ODE fails to reproduce important features of the dynamics, while the extended ODE, which is structurally aligned with the ABM, achieves strong agreement, see Figure 2. This reinforces the value of mechanistic fidelity when translating across modeling levels.

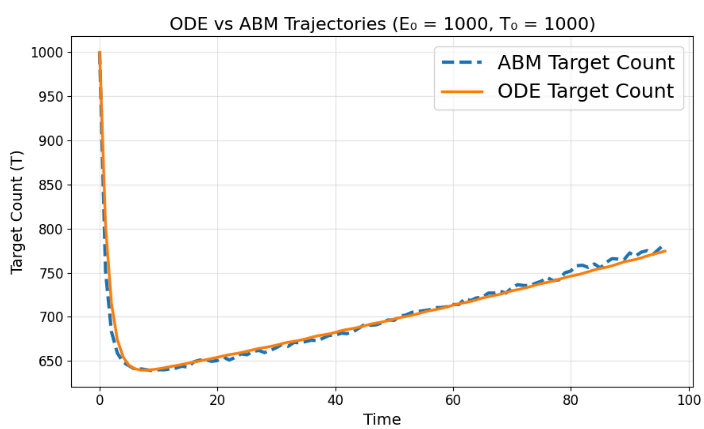


Figure 2: Fit extended ODE system to ABM synthetic data at one initial value pair

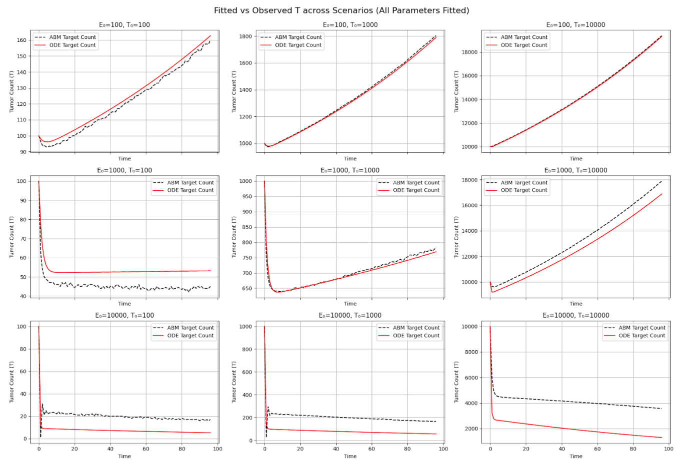


Figure 3: Globally fit extended ODE system to ABM synthetic data with different initial cell counts

One key difference lies in the interpretation of the binding rate. In the ABM, the binding probability refers to the chance that a given E-T pair within interaction distance binds. In contrast, the ODE binding rate reflects the overall likelihood of binding across the population, scaling with the product of E and T cell counts. As a result, the effective binding rate estimated from ODE fitting is substantially lower than the 0.4 used in the ABM. Other parameters, such as growth rate, unbinding rate, and killing rate, retain similar values across the two models.

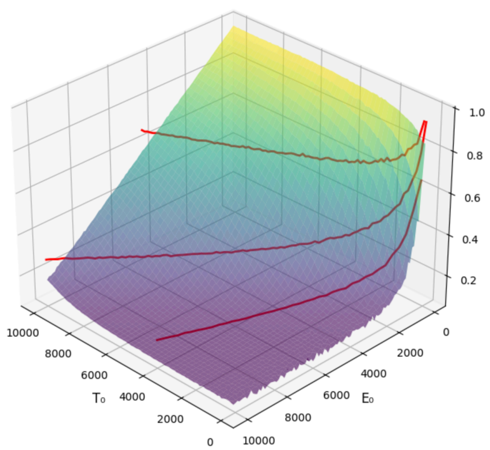
To further test the robustness of the ODE framework, we evaluate whether a single parameter set can fit all synthetic trajectories. When fitting the extended ODE model across the entire E-T grid, we find that a shared binding rate of 0.0003 offers a reasonably good fit across conditions (Figure 3). This suggests that a population-level model with consistent parameters can still approximate diverse spatial behaviors.

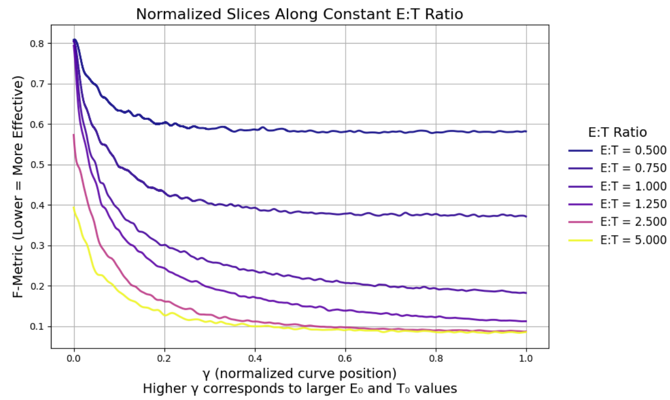
Both ABM and ODE models can be used to generate F-surfaces. Recall F is an effectiveness metric that lower F indicates more effective treatment. By slicing these surfaces along lines of constant E:T ratio, we obtain F-curves. Because the original curves differ in length due to the range of initial T values, we reparametrize them to a common horizontal scale for comparison.

Although the ABM and ODE generate slightly different surfaces, their F-curves align in both shape and relative position (Figure 4). Two important conclusions emerge:

- E:T ratio matters: higher E:T ratios consistently yield lower F values, meaning more effective treatment.
- Absolute cell count also matters: even at a fixed E:T ratio, having more total cells leads to better outcomes.

These trends can be interpreted mechanistically. In the ABM, higher cell counts increase the chances of E cells encountering T cells quickly, allowing treatment to begin earlier. With fewer initial cells, E cells spend more time searching, during which T cells may proliferate uncontrollably. In the ODE model, this effect is captured by the E×T term in the binding rate - larger initial values naturally produce stronger interactions and faster tumor killing.





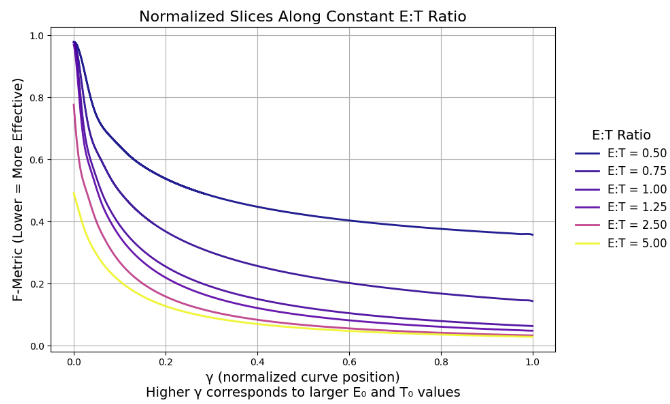


Figure 4: Left: F-surface with slicing demonstrated;
Top-right: F-curves using ABM;
Bottom-right: F-curves using ODE.

4.2. Fitting to Lab Data

We next apply our models to real experimental data. Unlike the synthetic trajectories generated by the ABM, the lab data is extremely limited: we are given only 18 endpoint measurements, each, after pre-process, consisting of an initial effector count, an initial target count, and the corresponding final target count after 24 hours. The data points span 3 different initial T values and 6 E:T ratios, thus forming a 3×6=18 grid of conditions.

To evaluate model performance, we fit both the ABM and ODE models by providing the starting E and T values and adjusting parameters to minimize the difference between the predicted and observed final T cell counts. In both cases, the models fail to capture the observed outcomes. The predicted and actual final counts differ significantly, and the overall fit is poor, see Figure 5.

The ODE fitting not only performs poorly in error metrics - it also produces biologically implausible parameter values. In particular, the best-fit solution often sets the unbinding rate to 1, effectively disabling killing altogether. This implies that the model favors uncontrolled tumor growth rather than attempting to reproduce cytotoxicity. This failure reflects a deeper structural mismatch between the model's assumptions and the observed data.

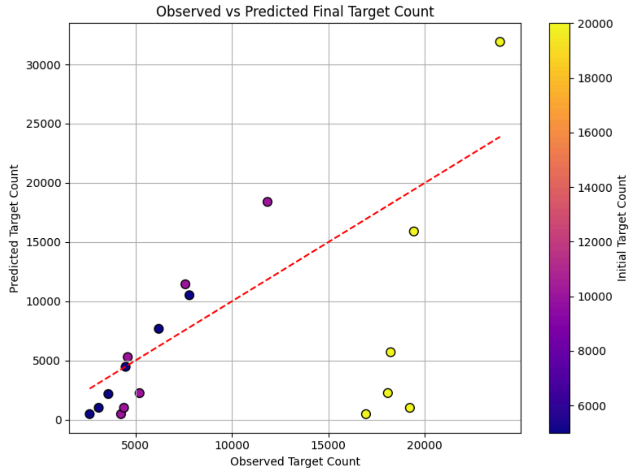


Figure 5: Observed lab data vs Predicted data

Closer inspection of the lab data reveals a pattern: within each group of fixed initial target count, only one or two points can be reasonably fit by the model, while others deviate substantially. This suggests that the mechanistic models overemphasize the role of the E:T ratio. Both the ABM and ODE assume that increasing E should consistently improve treatment effectiveness, but the lab data suggests diminishing returns - especially at high E:T ratios, where further increases in E lead to only marginal improvement.

A natural next step would be to refine the ABM and ODE structures to better reflect the biology observed in these assays. For instance, introducing saturation effects or effector exhaustion might reduce the sensitivity to E:T ratio. However, due to project constraints, we were unable to pursue these model adjustments in this work.

Instead, we turn to a simpler, data-driven alternative. We begin with linear regression to explore direct relationships between input variables and final outcomes, and ultimately adopt a Hill-type model to better capture the saturation-like behavior observed in the experimental data. This approach trades mechanistic interpretability for empirical flexibility and provides a useful baseline for comparison.

4.3 Hill Function Fitting

To better capture the relationship between effector dose and cytotoxicity in the limited lab dataset, we turn to a more empirical, saturation-aware model: the Hill equation. As a preliminary step, we examine whether a simple log-linear relationship exists between E:T ratio and cytotoxic effectiveness.

For each group of initial target cell count (5000, 10000, and 20000), we perform a linear regression using log-transformed E:T ratio as the independent variable and percent killing of tumor cells as the dependent variable. Each group contains six data points, and the goal is to determine how well E:T ratio alone explains variability in cytotoxic response.

The regression results are summarized below:

Initial Target Cell	Regression Type	R squared	Slope	Intercept	P-value	Std. Err.
5000	% Killing ~ log(E:T)	0.93	14.87	34.81	0.002	2.02
10000	% Killing ~ log(E:T)	0.71	9.66	56.19	0.035	3.09
20000	% Killing ~ log(E:T)	0.61	3.66	47.13	0.065	1.45

These results suggest that log(E:T) is a strong predictor of cytotoxicity when the initial tumor load is low. However, as the number of target cells increases, the regression fit weakens, indicating that E:T ratio alone cannot fully explain treatment effectiveness under high-burden conditions.

To address this limitation, we fit a Hill-type model to each initial T group. The Hill equation describes how response saturates with increasing dose, making it a natural candidate for modeling cytotoxicity across a range of effector levels. For each group (six points per group), we randomly select four for training and hold out two for testing.

Despite the small sample size, the Hill model fits the data well within each group, see Figure 6. It captures the saturation behavior and predicts test points accurately, suggesting that the Hill structure is well suited for these assays.

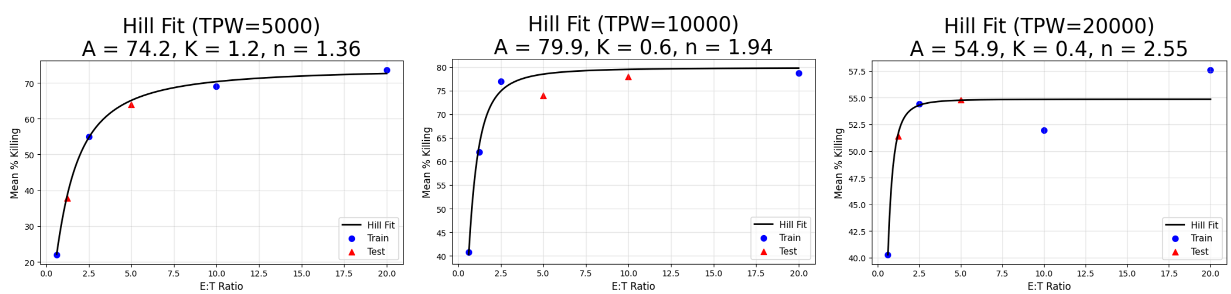


Figure 6: Fit and predict lab data to Hill Equation

This leads to two important conclusions:

- **E:T ratio is an important predictor**, but only within a fixed T cell background. The Hill model captures this relationship well.
- **Absolute tumor burden still matters**. A global Hill fit across all 18 data points performs poorly, see Figure 7, initial target count impacts the entire dose-response curve.

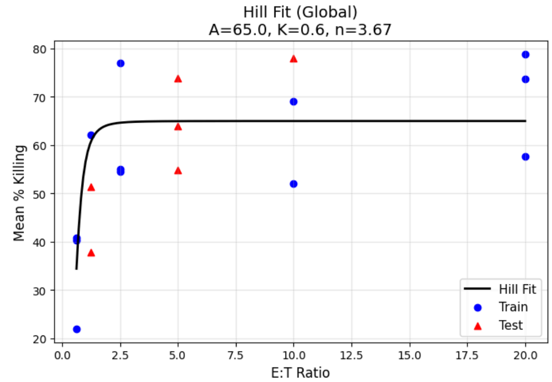


Figure 7: Globally fitting Hill equation

The fitted Hill parameters further support this interpretation, as initial target cell count increases:

- **A (maximum killing rate)** decreases: high tumor load diminishes overall efficacy.
- **K (half-max E:T ratio)** decreases: the system becomes more sensitive, needing less E:T to achieve half-effectiveness.
- **n (steepness, Hill coefficient)** increases: the response becomes more switch-like in high-density environments.

These trends offer a valuable direction for future work: modifying ABMs or ODEs to include saturation, limited killing capacity, or nonlinear dose sensitivity could help reconcile the gap between model prediction and experimental outcomes.

5. Additional Explorations

To bridge the gap between individual stochastic behavior and population-level averages, we also examined the master equation approach. This probabilistic framework describes the evolution of the full distribution of observing n target cells at time t. It explicitly accounts for cell proliferation and death as stochastic, constant-rate processes, making it particularly useful for systems with inherent biological noise.

For a set of wells with varying numbers of tumor cells, the master equation is written as:

$$\frac{dP_n(t)}{dt} = \lambda(n-1)P_{n-1}(t) + \mu(n+1)P_{n+1}(t) - (\lambda+\mu)nP_n(t),$$

where:

- The first term, $\lambda(n-1)P_{n-1}(t)$, accounts for transitions into state n due to a cell in state $n-1$ proliferating.

- The second term, $\mu(n+1)P_{n+1}(t)$, accounts for transitions into state n due to a cell in state $n+1$dying.
- The third term, $(\lambda+\mu)nP_n(t)$, accounts for transitions out of state n due to cells either proliferating or dying.

Boundary conditions depend on the specific setup (e.g., when n = 0, there is no death term since no cells exist to die, nor proliferate). We also have to be careful with n = 1, then the rest should follow the equation above:

$$\frac{dP_0(t)}{dt} = \mu P_1(t), \quad \frac{dP_1(t)}{dt} = 2\mu P_2(t) - (\lambda+\mu)P_1(t).$$

For the special case of a birth-death process with constant proliferation and death rates, the mean and variance of cell count at time t can be analytically computed as:

$$\langle n(t) \rangle = N_0 \cdot \exp((\lambda-\mu)t),$$
$$\mathrm{Var}(n(t)) = N_0 \cdot \exp\left(2(\lambda-\mu)t\right) \cdot \left(1 - \exp((\mu-\lambda)t)\right) \cdot \frac{\lambda+\mu}{\lambda-\mu}.$$

The Master Equation (ME) offers a useful lens for studying stochastic dynamics in cell populations. This framework is particularly relevant in settings where randomness plays a central role - such as early tumor growth, immune escape, or low-dose treatments. In principle, it can be extended to two or three dimensions to model joint dynamics of effector, target, and complex cells. In its 1D form, it tracks the full probability distribution of tumor cell counts over time, rather than just the mean, see Figure 8:

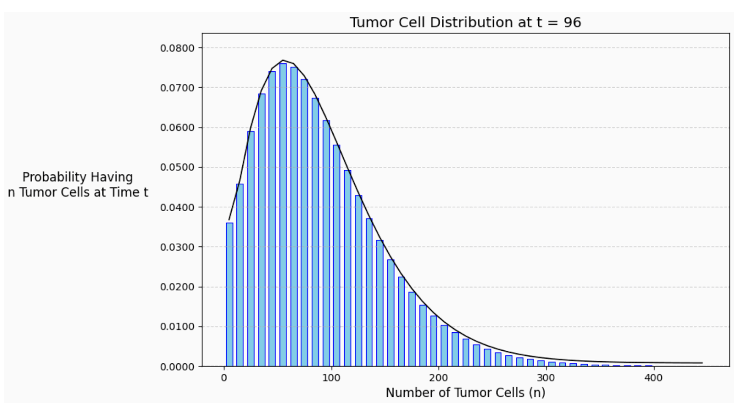


Figure 8: Master equation reveals distribution information

However, in our current setup, the ME provided limited added value. The resulting distributions were smooth and unimodal, offering little beyond what we already observed in ABM simulations. Extending to higher dimensions is also computationally intensive and was beyond the scope of this project.

Still, the ME remains a promising direction for future exploration, particularly for studying rare events, variability across replicates, or probabilistic treatment failure.

6. Conclusions and Ways Forward

Our project presents a multi-layered modeling framework that bridges controlled in vitro cytotoxicity assays with the complexity of immune responses in vivo. By integrating Agent-Based Models (ABMs), Ordinary Differential Equations (ODEs), and Master Equation (ME) methods, we explored key dynamics of T-cell dependent cytotoxicity (TDCC) in CAR T-cell and T-cell engager therapies.

A core insight from our study is that the effector-to-target (E:T) ratio alone does not fully explain treatment outcomes. Absolute cell numbers also play a critical role in shaping cytotoxic response.

Our ODE and ABM models capture core trends and provide a strong foundation for future refinement. While limited lab data posed challenges for fitting, these models remain biologically grounded and interpretable. In contrast, the Hill equation, though less mechanistic, offered a surprisingly strong fit and highlighted saturation and threshold effects in cytotoxic response. The ME approach offered valuable probabilistic insights, but was ultimately underutilized due to project scope.

Future work could focus on refining the ABM to better reflect cell exhaustion and contact limitations, extending ODEs to include spatial or delay effects, and using ME-based methods to capture early-stage variability. Collecting time-series experimental data would also greatly enhance parameter fitting and model validation.

Overall, our modeling framework provides a strong foundation for guiding experiment design, refining preclinical assays, and informing therapeutic development - offering Takeda actionable tools and insights for future TDCC studies.

7. References

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